

ICS2201 (Individual Assigned Practical Task)

Title:	Benchmarking Small-Object Detection on Brain MRI: A Cerebral Microbleed Case Study
Supervisor:	Charlie Abela
Co-supervisor:	
Main Subject Area:	Machine Learning, Medical Image Analysis
Short Description of the Task:	In medical imaging, detecting very small structures within MRI scans is a challenging problem that sits at the intersection of computer vision and clinical diagnostics. Cerebral microbleeds (CMBs) are tiny deposits of blood products in the brain, typically 2–10 mm in diameter, that appear as small dark spots on certain MRI sequences. Their accurate detection and counting is clinically important because microbleed burden is associated with stroke risk, cognitive decline, and cerebral small vessel disease. However, manually identifying these sparse, small objects across entire brain volumes is time-consuming and prone to inter-reader variability. The VALDO Challenge (MICCAI 2021) released a publicly available dataset of 72 subjects with T1, T2, and T2*-weighted brain MRI, together with expert annotations marking every microbleed. This provides a well-curated benchmark for developing and evaluating automated detection approaches. Modern self-configuring deep learning frameworks such as nnU-Net have shown strong performance on medical image segmentation tasks, but the specific challenge of detecting very small, sparse objects within large volumetric scans remains an open research question with important implications beyond just microbleeds. In this IAPT, the student will benchmark multiple detection and segmentation approaches on the VALDO cerebral microbleed dataset, systematically comparing their performance on this small-object detection task. The student will implement and evaluate at least two distinct approaches (e.g., nnU-Net 2D vs. 3D configurations, or a patch-based classifier vs. a full-volume segmentation model), analyse where and why different methods succeed or fail (e.g., by microbleed size, location, or proximity to vessels), and build an interactive visualisation dashboard that allows a user to explore detection results overlaid on the original MRI volumes alongside quantitative evaluation metrics. The final artefact will consist of the benchmarking pipeline, the trained models, the evaluation analysis, and the interactive dashboard, all accompanied by a written report documenting the methodology, results, and findings. The student is expected to deliver: a 10-minute presentation (15%), a video production demonstrating the artefact (15%), and the project artefact together with a written report (70%).

Recommended Resources (including readings):	VALDO Challenge (Cerebral Microbleed Detection): https://valdo.grand-challenge.org/ VALDO Training Data (Zenodo): https://zenodo.org/records/4520773 nnU-Net Framework: Isensee, F. et al. (2021). nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. Nature Methods, 18(2), 203–211. https://github.com/MIC-DKFZ/nnUNet VALDO Challenge Paper: Sudre, C.H. et al. (2024). Where is VALDO? VAscular Lesions Detection and segmentatiOn challenge at MICCAI 2021. Medical Image Analysis, 91, 103029. Small Object Detection in Medical Imaging: Liu, J. et al. (2021). A review of deep-learning-based medical image segmentation methods. Sustainability, 13(3), 1224. Interactive Medical Image Visualisation: 3D Slicer (https://www.slicer.org/) and NiiVue (https://github.com/niivue/niivue)
Prerequisite Knowledge Required:	Machine Learning, Deep Learning, Python Programming, Data Visualisation